AWS Auto Scaling Project

with Application Load Balancer

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Region:	Asia Pacific (Mumbai) — ap-south-1
Date:	June 16, 2026
Project Type:	Cloud Infrastructure — EC2 Auto Scaling

Project Architecture Overview

Component	Details
EC2 Instances	3x t2.micro (Running) — ap-south-1b
Auto Scaling Group	My-ASG | Min: 1, Desired: 1, Max: 3
Launch Template	AutoScalingTemplate (lt-0c8b0783a9f31cb9a)
Load Balancer	My-ALB (Application, Internet-facing)
Target Group	ASG-TargetGroup (HTTP:80, Instance type)
Security Group	ASG-SG (SSH:22, HTTP:80 inbound)
VPC	vpc-011cb6740d2593755

1. Project Objective

The goal of this project is to design and deploy a scalable, highly available web application infrastructure on Amazon Web Services (AWS). This involves setting up EC2 instances using an Auto Scaling Group (ASG) to automatically adjust capacity based on demand, combined with an Application Load Balancer (ALB) to distribute incoming HTTP traffic evenly across healthy instances.

Key Objectives:

- Configure a Security Group with appropriate inbound/outbound rules for SSH and HTTP access.
- Create a Launch Template to define EC2 instance configuration (AMI, instance type, user data).
- Deploy an Auto Scaling Group with a scaling range of 1 to 3 instances.
- Set up a Target Group and Application Load Balancer to route traffic across instances.
- Verify the setup by accessing the load balancer DNS and observing auto-scaling activity.
- Test automatic instance replacement when instances are manually stopped.

2. Security Group Configuration

A Security Group named **ASG-SG** was created to act as a virtual firewall for the EC2 instances. It controls inbound and outbound traffic as described below:

Rule Type	Protocol	Port	Source/Destination
Inbound — SSH	TCP	22	0.0.0.0/0 (Anywhere)
Inbound — HTTP	TCP	80	0.0.0.0/0 (Anywhere)
Outbound — All Traffic	All	All	0.0.0.0/0 (Anywhere)

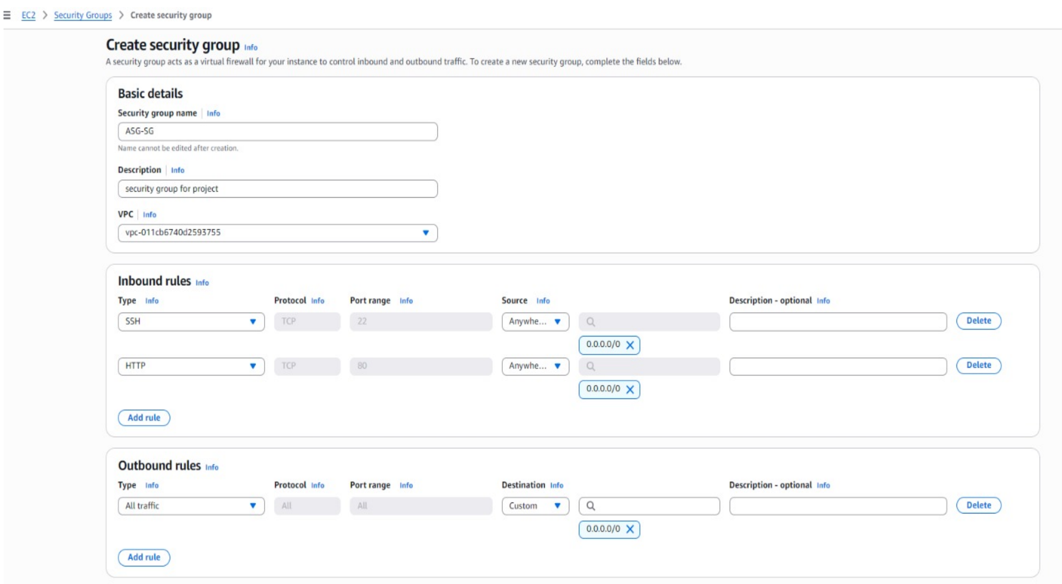

Figure 1: Security Group (ASG-SG) creation with inbound SSH and HTTP rules

3. Launch Template

A Launch Template named **AutoScalingTemplate** (ID: lt-0c8b0783a9f31cb9a) was created to define the configuration for EC2 instances launched by the Auto Scaling Group. The template specifies the AMI, instance type, security group, and user data script.

Parameter	Value
Template Name	AutoScalingTemplate
Template ID	lt-0c8b0783a9f31cb9a
AMI ID	ami-0e38835daf6b8a2b9
Instance Type	t2.micro
Security Group	ASG-SG
Default Version	1
Owner	arn:aws:iam::653893523299:root
Created On	2026-06-16T13:40:03.000Z

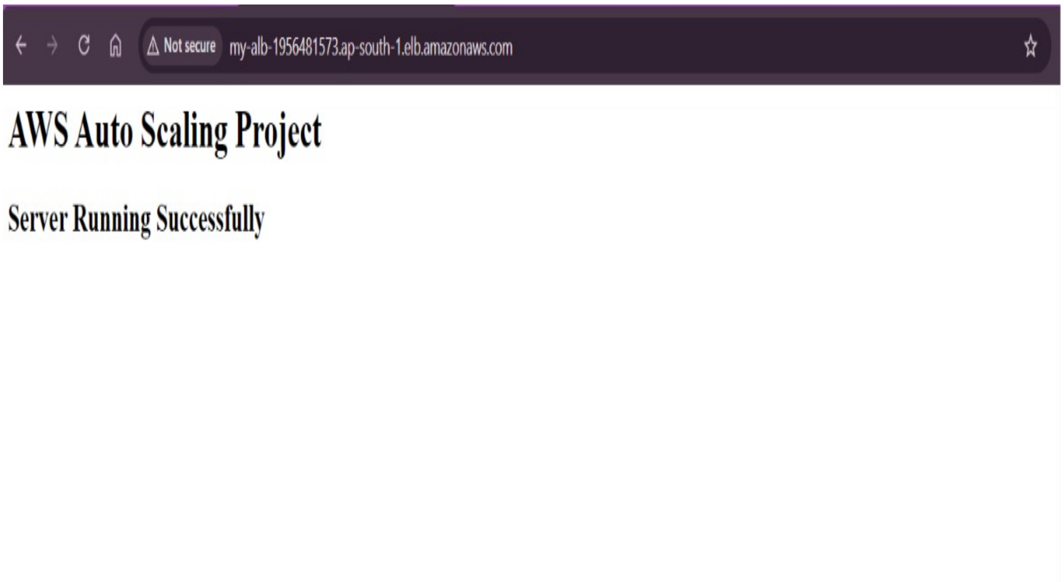

Figure 2: Launch Template (AutoScalingTemplate) with Actions menu showing available operations

4. Auto Scaling Group (ASG)

The Auto Scaling Group **My-ASG** was configured to automatically maintain the desired number of EC2 instances. It uses the AutoScalingTemplate and spans multiple Availability Zones for high availability.

Parameter	Value
ASG Name	My-ASG
ARN	arn:aws:autoscaling:ap-south-1:653893523299:autoScalingGroup:...
Launch Template	AutoScalingTemplate | Version Default
Desired Capacity	1 (initially), scaled to 2 during test
Minimum Capacity	1
Maximum Capacity	3
Availability Zones	ap-south-1a, ap-south-1b (2 AZs)
Status	At desired capacity
Date Created	Tue Jun 16 2026 19:33:59 GMT+0530

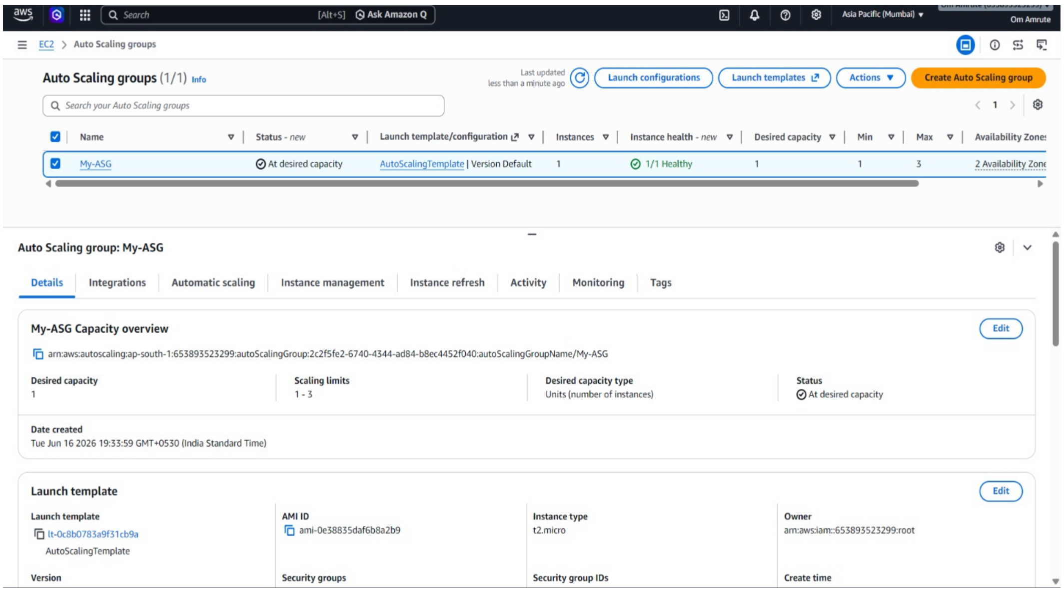

Figure 3: Auto Scaling Group (My-ASG) — Details tab showing capacity and launch template

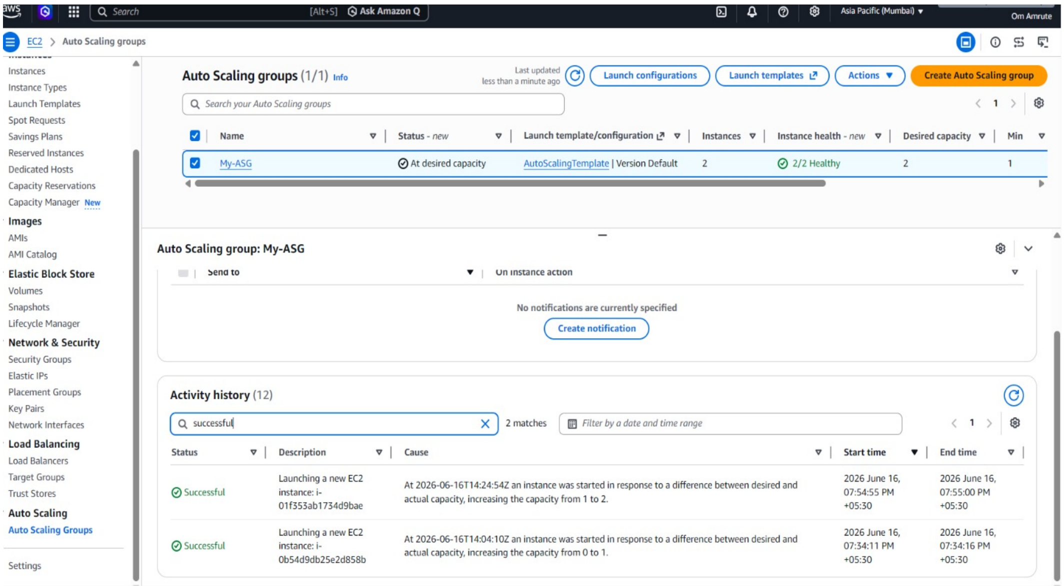

Figure 4: ASG Activity History — 2 successful instance launches recorded

5. Target Group

A Target Group named **ASG-TargetGroup** was created to route HTTP traffic to registered EC2 instances. The ALB forwards requests to this target group, which performs health checks and ensures only healthy instances receive traffic.

Parameter	Value
Target Group Name	ASG-TargetGroup
Target Type	Instance
Protocol : Port	HTTP : 80
Protocol Version	HTTP1
VPC	vpc-011cb6740d2593755
Health Status	2 Unused (pending ALB association at time of screenshot)
Registered Targets	2 instances (i-0628a87c4d7f39afc, i-07a2b59183a8e4775)

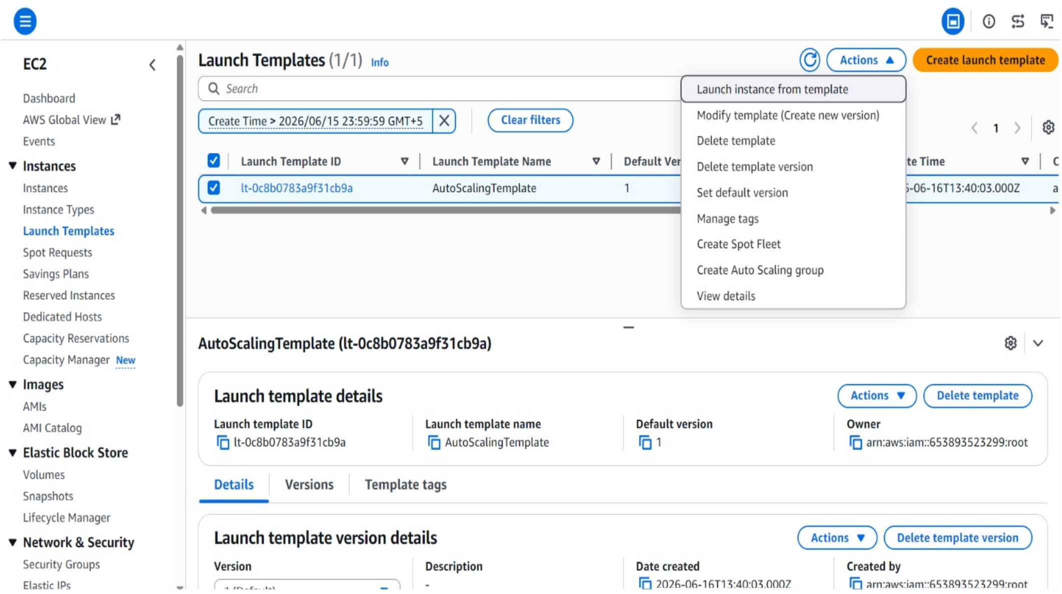

Figure 5: ASG-TargetGroup with 2 registered targets (HTTP:80)

6. Application Load Balancer (ALB)

An Application Load Balancer named **My-ALB** was created to distribute incoming HTTP traffic across EC2 instances registered in the ASG-TargetGroup. The ALB is internet-facing and listens on port 80.

Parameter	Value
ALB Name	My-ALB
Type	Application Load Balancer
Scheme	Internet-facing
IP Address Type	IPv4
DNS Name	My-ALB-1956481573.ap-south-1.elb.amazonaws.com
Hosted Zone	ZP97RAFLXTNZK
Listener	HTTP:80 → Forward to ASG-TargetGroup (1 rule, 100%)
Availability Zones	ap-south-1a (subnet-0efec95ee12e472df), ap-south-1b (subnet-0832363bce3ad4684)
VPC	vpc-011cb6740d2593755
Status	Provisioning → Active
Date Created	June 16, 2026, 19:25 (UTC+05:30)

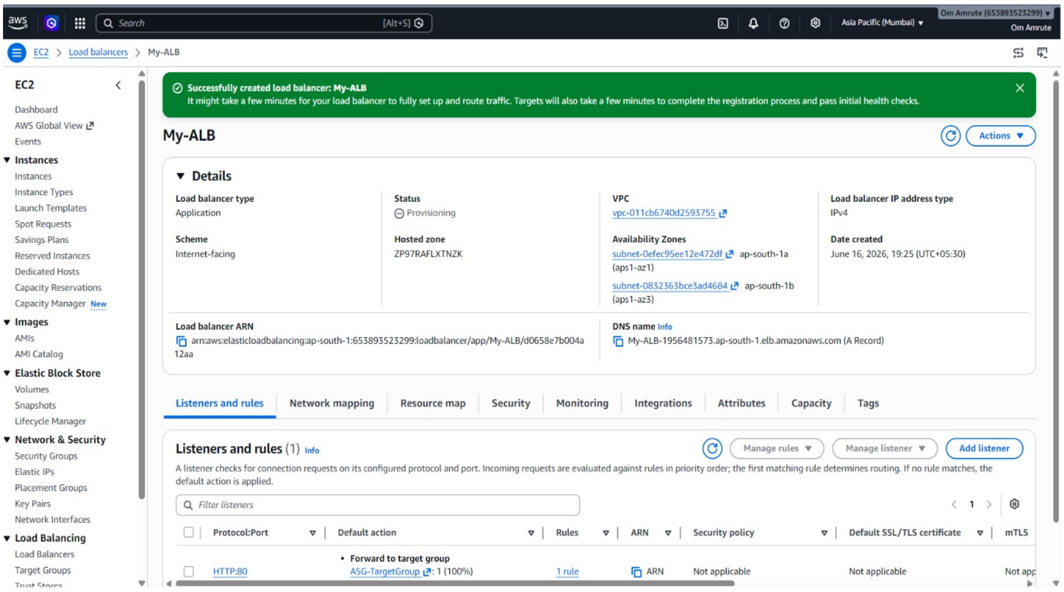

Figure 6: My-ALB successfully created with HTTP:80 listener forwarding to ASG-TargetGroup

7. EC2 Instances

After configuring the Auto Scaling Group and Load Balancer, **3 EC2 instances** were observed in the Running state in the ap-south-1b Availability Zone. All instances passed their status checks.

Instance ID	Type	State	Status Check	AZ	Public IP
i-0b54d9db25e2d858b	t2.micro	Running	2/2 passed	ap-south-1b	65.1.3.224
i-01f353ab1734d9bae	t2.micro	Running	2/2 passed	ap-south-1b	35.154.77.220
i-00c7f0c3cab43583d	t2.micro	Running	2/2 passed	ap-south-1b	13.233.101.217

Figure 7: EC2 Instances — 3 instances running (all passing 2/2 status checks)

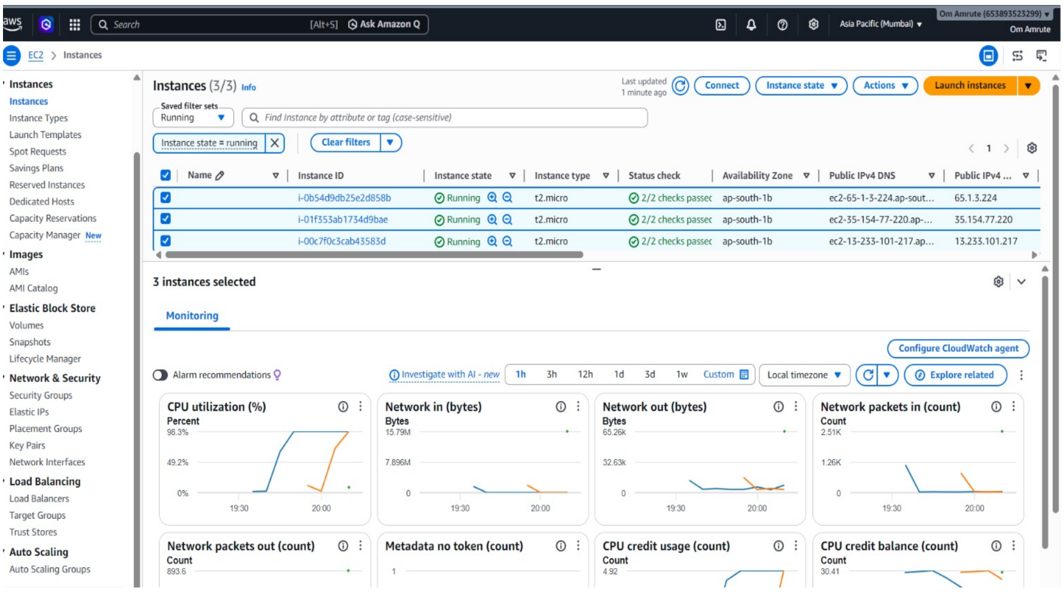

Figure 8: EC2 Monitoring — CPU Utilization reaching ~98.3%, confirming stress test / scaling trigger

8. Web Application Verification

After the ALB became active and targets were registered, the application was accessed via the ALB DNS name in a web browser. The server responded successfully, confirming that the load balancer is correctly routing traffic to the running EC2 instances.

URL Accessed: http://my-alb-1956481573.ap-south-1.elb.amazonaws.com

Response: "AWS Auto Scaling Project — Server Running Successfully"

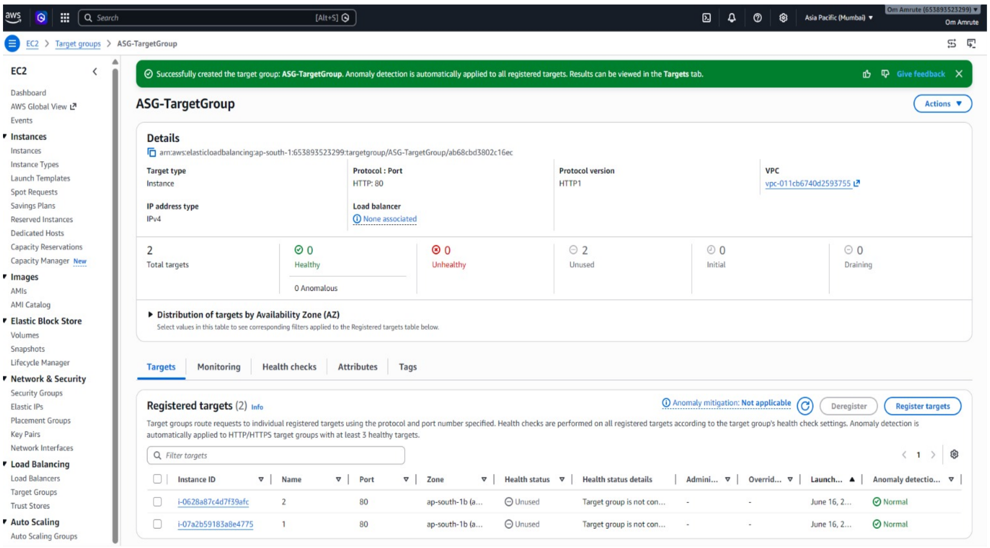

Figure 9: Browser output confirming the web server is reachable through the ALB DNS

9. Self-Healing / Auto Recovery Test

To verify the self-healing capability of the Auto Scaling Group, two instances were manually stopped. The ASG detected that the actual instance count dropped below the desired capacity and automatically launched replacement instances to restore the desired state.

Event	Description	Start Time	End Time	Status
Scale Out #1	Launched i-0b54d9db25e2d858b (capacity: 0 → 1)	2026-06-16 07:34:11 PM IST	2026-06-16 07:34:16 PM IST	Successful
Scale Out #2	Launched i-01f353ab1734d9bae (capacity: 1 → 2)	2026-06-16 07:54:55 PM IST	2026-06-16 07:55:00 PM IST	Successful

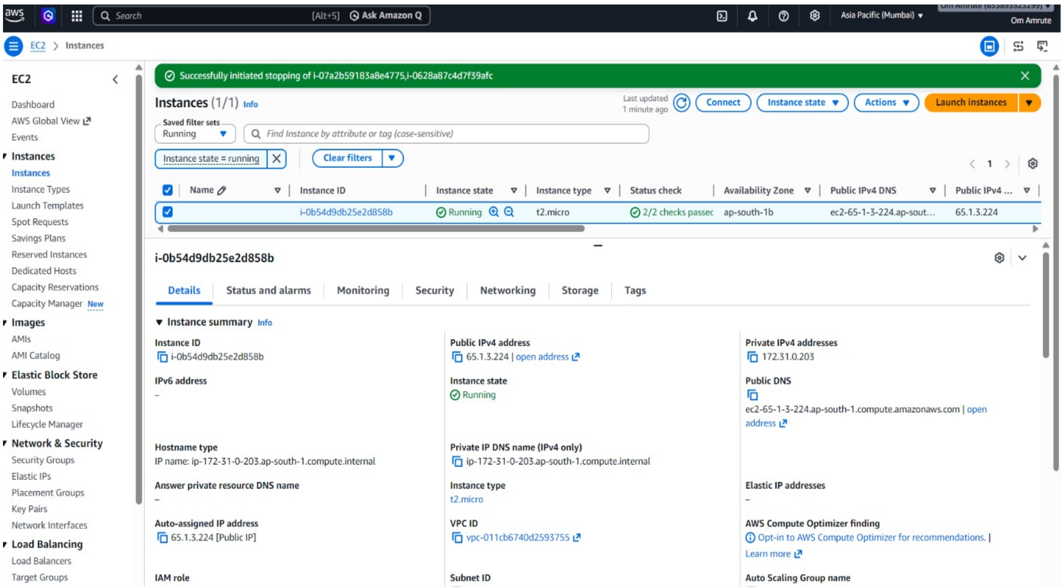

Figure 10: Instances i-07a2b59183a8e4775 and i-0628a87c4d7f39afc were manually stopped to trigger ASG recovery

10. Conclusion

This project successfully demonstrates the deployment of a scalable, fault-tolerant web application infrastructure on AWS. The key outcomes are summarized below:

- **Security Group:** Properly configured with SSH (22) and HTTP (80) inbound rules, enabling secure management and public web access.
- **Launch Template:** Standardizes EC2 instance configuration ensuring all auto-scaled instances are identical.
- **Auto Scaling Group:** Maintains instance count between 1 and 3, automatically scaling based on demand or instance failures.
- **Target Group:** Routes HTTP traffic to healthy registered instances with built-in health checking.
- **Application Load Balancer:** Distributes incoming web traffic across multiple instances and Availability Zones.
- **Self-Healing:** The ASG automatically detected and replaced stopped instances within seconds, confirming high availability.
- **Web Verification:** The application was successfully accessed via the ALB DNS, proving end-to-end connectivity.

The architecture built in this project is production-ready for stateless web applications. Adding HTTPS (SSL/TLS), custom scaling policies (CPU-based), and CloudWatch alarms would further enhance security and operational efficiency.

— End of Report —